

Nigeria electricity crisis: Power generation capacity expansion and environmental ramifications



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ARTICLE INFO

Article history:

Received 16 February 2013

Received in revised form

11 July 2013

Accepted 6 September 2013

Available online 4 October 2013

Keywords:

Energy crisis

Power sector reform

Nigeria

Vision 20:2020

Power generation

Environmental ramifications

ABSTRACT

Access to clean and stable electricity is essential in actualizing Nigeria's quest for joining the league of twenty most industrious nations by the year 2020 (vision 20:2020). No country can develop and sustain its development without having a minimum access to electricity for its larger percentage of its population. At present, Nigeria depends petroleum reserves and its aged hydro plant instalments for electricity generation to feed the 40% of its total population that are connected to the national grid. This paper summarizes literature on the current energy issues in Nigeria and introduces the difficulty of the issues involved. The paper also analyses the current (2010) electricity generation as well as the future expansion plans of the Government in 20 years period. The plan includes the introduction of new electrify generation technologies that have not been in used in the base year (2010). The electricity generation system of (including the future expansion plan) was simulated using the LEAP System (Long-range Energy Alternative and Planning). We also investigated the potential environmental impact of siting a nuclear power plant in one of the potential sites based on the site's specific micro-meteorology (land use) and meteorology using the US EPA (Environmental Protection Agency) models; AERMOD 12345.

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1. Introduction

Electricity was first generated for public use in Nigeria in 1896. Although it has been generated for over a century, electricity demand in Nigeria is at present far more than the supply, thereby affecting the country's socio-economic and technological developments [1,2].

Nigeria is the most populous country in Africa, with population of over 155 million people [3], and the majority of the citizens are living below the \$1.0 per day poverty level [4]. Only 40% of Nigeria's population is connected to the national electricity grid; the connected population faces power problems 60% of the time [1,5].

The energy crisis has crippled the industrial sector, which claimed it needed 2000 MW (e) to run in 2009, and the MAN (Manufacturers Association of Nigeria) says it spends more than N 1.8 billion (US \$ 11, 340 million) weekly in the running and maintenance of power generators [6]. The use of these generators in the industries has resulted in high cost of energy; since energy cost constitutes 40% of the production cost in Nigeria. At present, the cost of production in Nigeria is nine times higher than that of China [7].

The over-dependence of the energy sector on petroleum has slowed down the development of alternative fuels [8,9]. There is the need for diversification to achieve a wider energy supply mix, which will ensure greater energy security for the country.

Therefore, this paper presents an overview of the electricity crisis in Nigeria, the policy issues and environmental ramifications of the power sector reform act as well as dispersion modelling of discharges from Nigeria's (yet to be constructed) pioneer NPP (nuclear power plant). Within the context of this paper, clean energy technologies are the technologies that are reported to have lower CO₂ emission in literature. They are nuclear, wind, solar and hydro-electricity [10].

2. Potential energy sources in Nigeria

Nigeria is endowed with abundant natural energy resources, which includes crude oil, natural gas, coal and lignite, wind, solar radiation, Biomass and nuclear. It has the seventh largest natural gas reserve in the world and the largest in Africa.

In 2011, Nigeria produced about 2.53 million barrels per day (bbl/d) of total liquids, well below its oil production capacity of over 3 million bbl/d, due to production disruptions that have compromised portions of the country's oil for years. The economy of Nigeria is heavily dependent on its petroleum sector, which

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accounted for more than 95 percent of export earnings and more than 75 percent of federal government revenue in 2011. The untapped oil reserve of Nigeria is estimated at 37 billion barrels [11]. Aside this huge oil reserves, Nigeria holds the largest natural gas reserves in Africa, but has limited infrastructure in place to develop the sector. Natural gas that is associated with oil production is mostly flared, but the development of regional pipelines, the expansion of LNG (liquefied natural gas) infrastructure, and policies to ban gas flaring are expected to accelerate growth in the sector, both for export and domestic use in electricity generation. Uncertainties in Nigeria's investment policies and regulatory framework have caused a slowdown in oil and gas exploration activity, and delays in project development, including LNG projects. The natural gas reserve of Nigeria was estimated at 187.44 Tcf as of 2005. Other fossil fuel reserves of Nigeria are coal and lignite, tar sand, fuel wood and renewable sources like hydro, solar and wind.

From 2003 till date the US has been the major buyer of Nigeria's oil. The US Energy Information Administration [11] estimated that the U.S. has imported between 9 and 11 percent of its crude oil from Nigeria; however, U.S. import data for the first half of 2012 show that Nigerian crude is down to a 5 percent share of total U.S. crude import. Fig. 1 presents the crude oil production and consumption in Nigeria, between the years 1995 and 2011. The average daily production in 2011 was 2.15 million bbl/d. The declaration and implementation of the amnesty program between the years 2009–2010 have yielded a positive result, as the oil production capacity increases due to reduction in the number of attacks on pipelines and oil theft in the Niger delta region.

Table 1 shows a breakdown of the energy reserves and energy potentials in Nigeria. It is obvious that Nigeria has enough resources to cater for its energy need. Some of the resources are not tapped; the potential is vital for Nigeria's economic growth, but the access and utilization, which are the major drivers of the growth, are lacking [12].

Although Nigeria coal reserve is estimated to be 40 billion tons, the fuel (coal) has not been in full used for electricity generation in Nigeria. More attention is given to oil and gas as well as hydro-power for electricity generation. Hydro (as a renewable source) has been playing a substantial role in the Nigerian power sector. On a global scale, less than 15% of primary energy supply is renewable energy, and the major part is wood fuel and hydro power in developing countries [13]; and worldwide, the later and wind power are predicted to provide the largest share of the projected growth in total renewable generation [14].

Nigeria generates electricity at a commercial scale from four major sources: natural gas, oil, hydro and coal. Fig. 2 presents the percentage contributions of each of the sources. Since coal is neglected, petroleum (oil and gas) has contributed over 70% of the commercial primary energy in Nigeria [8].

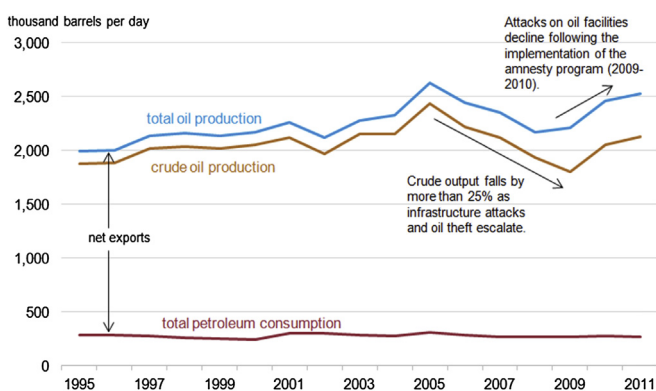


Fig. 1. Oil production and consumption in Nigeria, 1995–2011 [11].

Table 1
Nigeria's energy reserve per capacity as at December 2005 [4,12].

Energy source	Reserves
Crude oil	36.5 billion barrels
Natural gas	187.44 Tcf
Tar sands	30 billion barrels of oil equivalent
Coal and lignite	Over 40 billion tons
Large hydro	11,235 MW
Small hydro	3500 MW
Fuel wood	13,071,464 ha
Animal waste	61 million tons/yr
Crop residue	83 million tons/yr
Solar radiation	3.5–7.5 kwh/m ² -day
Wind	2–4 m/s at 10 m height
Wave and tidal energy	150,000 TJ/(16.6 × 10 ⁶ toe/yr)

The over-dependence of the Nigerian energy sector on petroleum has slowed down the development of alternative fuels. In order to achieve the Vision 20:2020, efforts must be made toward achieving a diversified energy supply mix, which will ensure greater energy security for Nigeria.

3. Nigeria's electricity power sector outlook

The NESCO (Nigeria Electricity Supply Company) commenced operations in 1929, in the attempt to connect all parts of the country and ensure secured electricity supply, NESCO has undergone so many transformation and reforms. It was renamed NEPA (National Electric Power Authority) in 1972. NEPA was known to have a burden of subsidies, low service quality and woeful collection of tariff. The reform act of 2005 unbundled NEPA into 18 companies (under the flag of Power holding Company of Nigeria): 6 generation companies, 1 transmission company and 11 distribution companies. The generating companies are made of 3 hydro and 9 thermal (gas based) stations with their output shown in Table 2 [15].

The structure of the PHCN (Power Holding Company of Nigeria) was to oversee the activities of the Managing Director/CEOs of the successor company for a period of 5 years, which should be adequate time to enable the companies to be privatized [2]. The privation exercise has suffered many drawbacks and its objectives have not been achieved. Recently, the minister of power who was part of the privatization committee resigned from office, citing gross corruption and sabotage as his reasons for leaving the office.

3.1. Nigerian grid system

The available capacity is about 49% of the installed capacity (Table 2), and recorded available capacity in recent years is about 4300 MW. As of November 2011, the average power generation in Nigeria was 3200 MW.

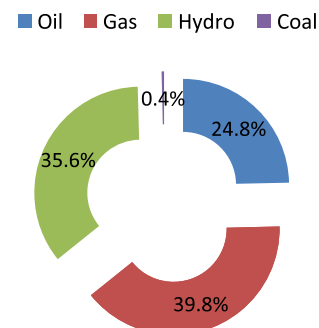


Fig. 2. Percentage contribution for the energy sources in Nigeria [8].

Table 2
Nigeria's power generating plants and their capacity utilization [16,17].

Power station	Type	No. of units	Year of construction	Age (yrs)	Installed capacity (MW)	2011 Available capacity (MW)	% Contribution to the national grid (approximate)
Kianji	Hydro	12	1968	43	760	480	14.3
Jebba	Hydro	6	1985	26	540	450	13.4
Shiroro	Hydro	6	1989	22	600	450	13.4
Egbin	Thermal	6	1986	25	1320	1100	32.8
Geregu	Thermal	3	2007	4	414	276	8.2
Omotosho	Thermal	—	2007	4	304	76	2.3
Olorunsogo	Thermal	—	2008	3	304	76	2.3
Delta	Thermal	20	1966	45	900	300	8.9
Sapele	Thermal	10	1978	33	1020	90	2.7
Afam	Thermal	18	1963	48	726	60	1.8
Calabar	Thermal	—	1934	77	6.6	Nil	Nil
Orji river	Thermal	4	1950	61	10	Nil	Nil
TOTAL					6904.6	3358	

The reasons for the underutilization of the electric power plants are: lack of trained manpower, lack of local manufacturing capabilities of spare parts, weak transmission and distribution infrastructure, poor utility performance and theft of grid equipment as most of the equipment are located in unsecured places. Consumer indiscipline, low tariff collection rate, extraordinary transmission losses, insufficient funding, as available investment goes into replacement of distribution equipment instead of grid's expansion are also part of the reasons for the underutilization of the grid. Almost 80% of the internally generated revenue by the PHCN is used for payment staff salary and welfare packages [15].

The underutilization of the plants can be clearly shown in Fig. 3, which was extracted from the 2011 Presidential Retreat on Power. The figure also shows that the overall performance of the FGN (federal government of Nigeria) hydroelectric power plants is relatively better, if compared with the thermal plants based on the ratio of available to installed capacity.

Another major reason for the underutilization of the plants is their age [2]. As power plants get older the capacity factor of plant decreases due to lack of proper routine maintenance and security of public properties and general corrupt practices within the sector (which are cases peculiar to Nigeria). The capacity factor is a measure of the plants operating efficiency. It is defined as the total amount of energy produced during a period of time divided by the amount of energy that the plant would produce at a full capacity [14].

Limited funding and lack of accountability are also part problems the Nigerian power sector is facing. In the last 40 years (see Fig. 4), substantial investments into the power sector were

witnessed during the civilian government, which came into power in May 1999.

The sector experienced insufficient funding between the early and late 90s. This translates into the present power outages the country is faced with. During the last 12 years of democracy, the government has made energy security and supply one of its top priorities and that can be seen clearly Fig. 4. It is important to mention that relatively good investments were made into the power sector between the late 70s and early 80s and during this period, the population was smaller.

If the 2005 power sector reform act was fully implemented, that should have allowed for the deregulation of the energy sector thereby reducing the fiscal burden of financing the sector on the federal government, which has been the sole financier of the sector since after independence in 1960. The FGN involvement in the power sector might have followed the political history of the country where the military government which has dominated the administration, restricted the concept of developing the power sector as an exclusive responsibility of the FGN.

A major policy review by the FGN was the licensing of 20 IPPs (independent power producers) in 2011 whose target is to generate additional 6258 MW within the first 3 years of their operation. One of these IPPs is the Zuma Energy Nigeria Limited, which is to generate 1200 MW from coal deposits in Itobe, Kogi state. The FGN also formally warranted state governments in the country to generate and distribute electricity within their territories [20].

4. Energy consumption pattern in Nigeria

The demand of electricity in Nigeria's outreaches the supply. About 60% of the population – over 80 million people are not served electricity with the rural and semi urban access to electricity estimated to be 35%. The poor performance of the electricity power sector is uncommon with countries that adopted the sectorial model in their power industries [12,21].

Fig. 5 shows that more than 6% of households have supported their access to electricity with power generator sets and about 3% of the households relied completely on the power generator sets for electricity. The rural electrification program has not made any significant contribution since only 1.1% of the rural households have been connected to the national grid through the program. The irony is that a lot of funds are been budgeted by the state and federal Governments for the rural electrification program almost every year, but this has not yield any significant contribution to the electricity supply in the rural areas. The failure of the program, among other issues is the measure factor that encourages rural urban migration in the country, which is a key factor that contributes to the increasing crime rate in the urban areas.

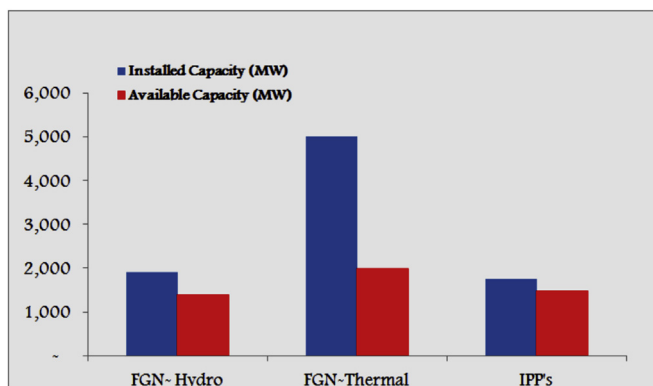


Fig. 3. Installed Vs. available Capacity of power generation by type in Nigeria (IPPs stands for Independent Power Producers) [18].

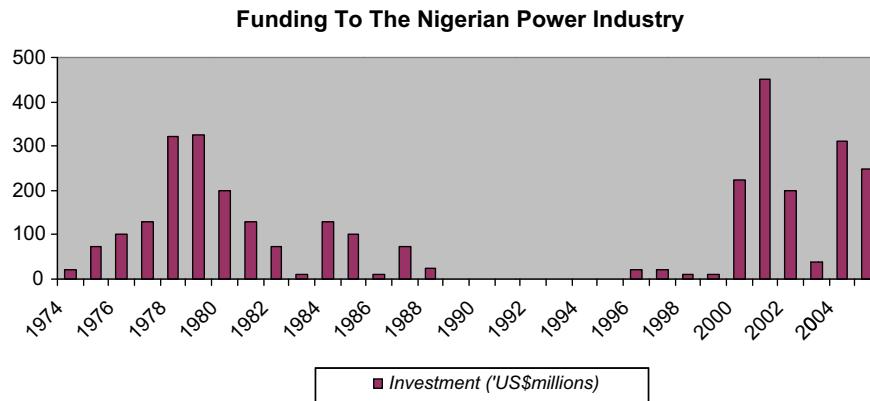


Fig. 4. Funding of the Nigeria Power sector in the last 3 decades [18,19].

The per capita electricity consumption in Nigeria is estimated to be 125 kWh against 4500 kWh and 1934 kWh in South Africa and Brazil respectively. In 2008, it was observed that more than 6% of households have supported their access to electricity with power generator sets and about 3% of the households relied completely on the power generator sets for electricity. These figures show a 6.6% increase in the number of households without electricity between the years 2007 and 2008 without adequate increase in the electricity generation capacity in the country. This has made the people cater for themselves by using other energy sources without considering the health risk associated with their use.

A survey conducted in Ref. [22] showed that only 6.2% of the firms in the Nigeria manufacturing sector exclusively rely on the grid electricity. Virtually all the firms (sampled for survey) have independent power generating set to support their activities. Table 3 shows the contribution of power outages to capacity underutilization in the manufacturing sector. In this survey, 44% of the firms attributed a fall of between 20% and 49% in their capacity underutilization to power outages. Twenty-four percent of small-scale firms lost over 50% of their output to unstable electricity compared to 14% and 17% for the medium and large enterprises, respectively.

The ECN (Energy Commission of Nigeria) computed the electricity demand projections with drivers of electricity demand

namely demography; socio-economy and technology the result of the projection are presented in Table 4 [2,15].

The supply outlook was computed using MESSAGE (Model for Energy Supply Strategy Alternatives and their General Environmental Impact), which takes into account demand variations of different final energy over the period as well as the different technologies and public policy on energy supply.

It can be deduced from Tables 4 and 5 that in 2010, (7% GDP growth) the projected demand was 15730 MW and the supply was 15668 MW. Going by the presidential pronouncement (13% GDP growth), the demand rose from 5746 MW in the base year to 297,900 MW in the year 2030 which translates to construction of 11,686 MW every year to meet the demand. The corresponding cumulative investment cost for the 25-year period is US\$ 484.62 billion, which means investing US\$ 80.77 billion every five years within the period [24].

The national load forecast shows that the load was about 10.5 GW by 2010 and that will triple by 2020; and the Government plans to improve the generation capacity to about 40 GW in the same year in order to achieve the Vision 20:2020, this contradicts the Energy Commission of Nigeria estimate for Nigeria to achieve the Vision 20:2020 with its fluctuating 3000 MW available capacity, the country needs to invest US\$ 150 billion to generate the 100,000 MW required for full industrialization. Achieving 100 GW is a very unlikely projection, probably based on a politician's promise. In fact, extremely low installed capacity is sufficient to satisfy the current needs. It is clear that 0.018 kW/person of installed power is an unacceptably low value, but to reach a value thirty-fold in 20 or so years appears absolutely out of the question. A detailed forecast analysis of the national electricity demand forecast (funded by the World Bank) [25] shows how the demand will change in the future. The forecast has been adopted by the current study for the simulation of the Nigeria electricity generation system using the current installed and the future capacity expansion plans.

5. Nigeria's electricity expansion plan

The FGN's power expansion plans indicate that the power sector will undergo a significant change within the short to medium time

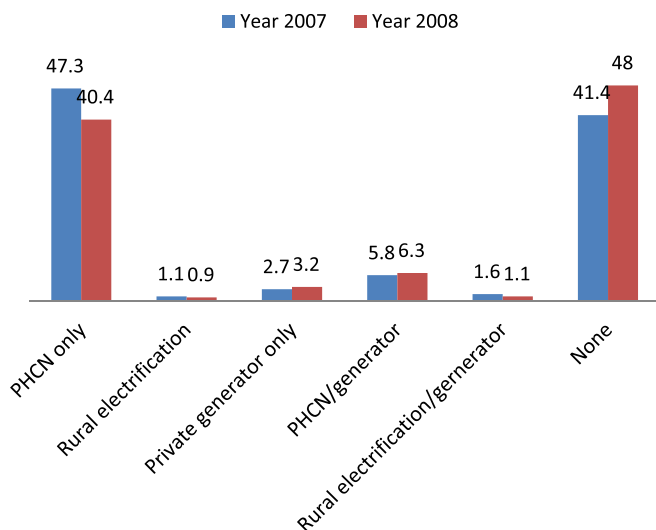


Fig. 5. Percentage distribution of electricity supply in 2007 and 2008 [12].

Table 3
Contribution of power outages to capacity underutilization.

Scale	Output loss as % of total output	% Contribution of outages to capacity underutilization
Small-scale firms	24	51.5
Medium-scale firms	14	36.7
Large-scale firms	17	39.2

Table 4
Electricity demand projection per scenario (MW) [2,15,23].

Scenario	2005	2010	2015	2020	2025	2030
Reference (7% GDP)	5746	15,730	28,360	50,820	77,540	119,200
High-growth (10%)	5746	15,920	30,210	58,180	107,220	119,200
Optimistic I (11.5%)	5746	16,000	31,240	70,760	137,370	250,000
Optimistic II (13%)	5746	33250	64,200	107,600	172,900	297,900

period. From the FGN's proposal, the generation capacity of the grid is set to increase by almost four times of the installed capacity by 2030 with the IPPs expected to play vital roles in the plan [26].

In its desperate attempt to address the energy poverty, the Government may consider solely the further development of conventional electricity technologies (like coal, oil and gas) that are readily available in Nigeria with little or no concern on the environmental impact of these technologies. As the world is moving towards an agreement that would charge power plants for CO₂ emission (due to the increasing threat of global warming), the days of cheap electricity from the conventional technologies will be gone if emission charges are included [27]. The situation of the Nigerian electricity consumers is disturbing such that the environmental issues may not be for now of significance among the public. Nigeria's CO₂ emission was estimated to be 36.9 million tons in 1985, and on the assumption that no gas was flared in 2025, this figure was estimated to rise to 73.6 million tons [28]. This is an indication that the country should consider clean technologies in curtailing its energy crisis.

The current and future capacity mix of Nigeria is presented in Table 6 and Fig. 6. Base on the power sector reform [29] and the national implementation plan of vision 20:2020 [30]. Gas and hydro fuels will maintain their positions as the main drivers of the electricity sector in the short and medium term. Renewable fuels like solar, Biomass and wind are expected to play roles in sustaining the Vision 20:2020; though their full potentials are not going to be tapped. This shows that the economy of Nigeria will be reliant on its fossil reserve for a longer period of time. The generation capacity will grow from the 6.9 in the base year to over 25 GW.

The future current (2010) and future (up to 2030) (Fig. 6) energy mix shows the government's plan to diversify the country's energy mix by expanding the fuel types, which include oil, gas, coal, nuclear, wind and solar. This will reduce the over-dependence of the power sector on petroleum, which has slowed down the development of other fuels that are available in Nigeria. The hydro power capacity is expected to increase from 1300 MW in the base year to about 5800 MW in the end year. The capacity of the gas (thermal) plant will increase from 5600 MW to 13,600 MW by 2030. The coal capacity is expected change from almost nil to 1300 MW by 2013. Nuclear energy is expected to generate 1000 MW by 2022 and the capacity is expected to grow by threefold of the base year value in 2030. The solar capacity is expected to increase from 75 MW by 2020 to 475 MW by 2030.

The current energy policy is critical to tackling carbon emission, which causes climate change and emphasises the government's willingness to pursue nuclear energy in full capacity [31]. The policy deemphasises the use of fuel wood as part of the country's

Table 5
Electricity supply projections per scenario (MW).

Scenario	2005	2010	2015	2020	2025	2030
Reference (7% GDP)	6440	15,668	28,356	50,817	77,450	113,6879
High-growth (10%)	6440	15,861	30,531	54,275	107,217	119,279
Optimistic I (11.5%)	6440	15,998	31,235	71,964	177,371	276,229

Table 6
Current and future electricity mix in Nigeria [1,26,28].

Technology type	Capacity (MW) 2003	Additional capacity (MW) 2010	Additional capacity (MW) 2020	Additional capacity (MW) 2030
Gas	4520	6901	2284	1320
Coal	n/a	388	n/a	1320
Oil	32	n/a	n/a	n/a
Hydro	1920	n/a	4740	5748
Biomass	n/a	n/a	5	5
Wind	n/a	n/a	20	20
Solar PV	n/a	n/a	75	425
Solar thermal	n/a	n/a	1	20
Nuclear	n/a	n/a	1000	4000
Total addition		7289	8280	12,858
Cumulative total	6472	13,761	20,276	29,394

energy mix, as it encourages deforestation and contribute heavily to the country's high CO₂ emission.

The environmental consequences of siting and operating an energy facility are enormous, as the facilities may lead to disruption of the ecosystem. On the other hand, any expansion on Nigeria's grid will reduce the use of private generators, which tend to be more environmental damaging as well as sources for noise pollution; the diesel-fuelled generators emit a complex mixture of air pollutants, which are responsible for chronic respiratory diseases and lung cancer in non-smokers [32].

Hydropower is one of the outstanding sources of base load electricity in Nigeria. Although it has high initial capital cost, hydropower is a cheap and clean electricity source. Nigeria has some inland rivers with great hydropower potentials. Rivers Niger and Benue and their tributaries constitute the core of the Nigerian river system, which offers a renewable source of energy for large-scale (greater than 100 MW) hydropower development. In addition, several scores of small rivers and streams do exist and can be harnessed for small-scale (less than 10 MW) hydropower projects [33].

The total exploitable large-scale hydropower potential of the country was estimated at over 10,000 MW, capable of producing 36,000 GWh annually. Table 7 indicates that only about one-fifth of hydro-electricity potential is developed. The small-scale hydropower potential is estimated at 734 MW. The small hydropower plants can be developed for the provision of electricity for the rural areas and remote settlements.

Low performance of the large hydropower plants is due to insufficiency of water in the Niger River, which powers the Jebba,

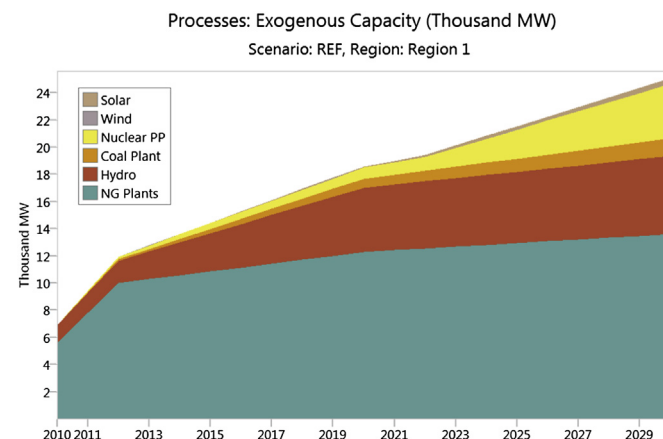


Fig. 6. Current and future committed capacity of Nigeria to achieve and sustain the Vision 20:2020.

Table 7
Small hydro scheme in Nigeria [34].

Plant location	Installed capacity (MW)	Available capacity (MW)	Available factor %
Bagel I	1	0.88	88
Bagel II	2	1.95	97.5
Ouree II	2	1.94	97
Kurra	8	7.5	93.75
Lere I	4	3.8	95
Lere II	4	3.3	95
Bakalori	3	2.87	95.67
Tiga	6	5.77	96.16
Total	30	28.51	94.76

Shiroro and Kijiji hydropower plants, and the efforts of the past administration to dredge the river is been faced by much political opposition. Another issue of great concern in the further development of hydro power in Nigeria is the post-construction effects of the Cameroonian Lagdo Dam on the River Benue. This has resulted in the reduction of water volume in the Benue River and the overflow of water from the dam often results in flooding around the riverine communities [34]. The River Niger dredging project would improve the water requirements in the inland drainage system as well as their tributaries for further exploitation of the hydropower potentials.

In addition to the large hydropower plants in Nigeria (Kijiji, Jebba and Shiroro), with combined installed capacity of 1900 MW, there are eight off-grid hydropower schemes in Nigeria. According to the FGN, electric power expansion plan (Table 6), hydropower is hope to contribute about 27% of the needed amount of electricity (25,000 MW) for the actualization of the Vision 20:2020.

Currently there are over 52 potential sites for the development of small hydro power plant in Nigeria. The development of hydropower is associated with environmental and social issues of concern; as construction of dams has led to loss of farmlands, displacement of rural settlement, and alteration of ecosystem of the host environment among other negative impacts. The practice in Nigeria is to compensate displaced settlements; like the case of Gurara dam where over 100 settlements were displaced. Although compensation was made, the affected people lost their farmlands, cultural artifacts, ancestral homes and socio-economic mainstays [34].

However, the environmental impacts of fossil-fuelled technologies are of greater regional and global significance than those of hydropower. These impacts include the effects of extraction and transportation of fossil fuels, emissions of acid-producing compounds, emissions of GHG (greenhouse gases), and disposal of large volumes of solid waste [35].

Wind is associated with little or no carbon emission. The World Nuclear Association's analysis [10] on life cycle GHG emissions of different energy technology shows that wind is the cleanest energy technology. The availability of wind for the production of commercially viable electricity in Nigeria is limited and varies with seasons. The needed speed of wind for the production of commercial electricity is 7 ms^{-1} [36]. This makes it unsuitable for large-scale production of electricity in Nigeria, because the maximum wind speed recorded over the period of 10 years at 10 m height is between 3.78 and 3.98 ms^{-1} around the northeast and northwest regions of the country [8] and the average wind speed across Nigeria is about 3 ms^{-1} [37].

Although the average wind speed across Nigeria is not economically viable for commercial scale electricity generation, the wind technology can be considered for water pumping, rural electrification and other forms of decentralized electricity generation in rural settlements of Sokoto, Guasau, and Kano (Fig. 7) where

the average annual wind speed is higher and suitable for these purposes [8,37,38]. Going by the electricity expansion plan of the country (Table 8), wind will contribute 0.08% of the total electricity needed for the actualization of the Vision 20:2020. The contribution of the wind energy could be further improved as funds are further expended in research and development of the wind technology.

Solar could be used for providing electricity to small settlements that are not connected to the national grid in some parts of the northern regions where studies have indicated that the monthly solar radiation potential ranged from 7 to $5.6 \text{ KW h/m}^2 \text{ days}$ [39]. Other applications of solar that could be expanded in northern Nigeria are water pumping, traffic lighting, rural clinic and school power. The power sector expansion plan (Table 6) indicates that solar technology is hope to contribute about 0.3% of the required amount of electricity that is needed for the actualization of the Vision 20:2020. The technology will contribute more in the sustenance of the vision 20:2020 than wind and Biomass. This is as a result of the increased solar energy capacity (of 475 MW) by 2030; this is almost a 600% increase in the solar capacity between 2020 and 2030.

Another option Nigeria seeks to adopt in curtailing its energy crisis is the option of Biomass energy, which has the very important benefits of contributing to security of fuel supply, support for agriculture and low GHG emissions. Little investment has been made in R&D in the area of biofuels in Nigeria, as more attention is given to the country's hydrocarbon reserve which is the major driver of the economy. Aside the issue of the funding R&D in biomass energy, this technology may be subjected to public criticism in the African most populous nation where the food production is not sufficient enough to cater for its domestic demand. In the same vein, there is strong concern about the effects of bioenergy on food production cost: in the last few years the price of corn in the US has doubled because of its increasing use for ethanol production [40].

5.1. Simulation of the Nigeria electricity generation system

To simulate the electricity generation system in Nigeria, we considered the total generation output of the various electricity generation technologies in the country in 2010 (base year). The total electricity demand forecasts of three different scenarios (SC1, SC2, and SC3) were also considered. These scenarios represent 3 different growth rate projections of the national peak load between the years 2010 (base year) and 2030 (end year). SC1, SC2 and SC3 represent 6.8%, 8.9% and 11.6% growth rate of the national peak load respectively. In the base year, the total electricity demand in Nigeria was estimated 17,675 GWh and the demand was estimated to grow to 100,000, 160,000 and 300,000 GWh in the SC1, SC2 and SC3, respectively. The total electricity demand of SC1, SC2 and SC3 are part of the revised national load forecast [25].

The current and future committed capacity of electricity generation in Nigeria reported in [25,29,30] as well as Table 7 was adopted as the input data for the simulation of the Nigeria electricity generation system using the Long-range Energy Alternatives Planning (LEAP) system [41]. The electricity demand of SC1, SC2, and SC3 were considered using the module Demand: Technology with total energy in order to give the power plants reasons to run in the reference scenario. The electricity losses decreased from 17% in the base year to 12% in the end year based on the FGN's plan [29]. The FGN also plans to increase the access to electricity from 40% to 50% of the population within the simulation period.

The results of the analysis for SC1, SC2 and SC3 are presented in Figs. 9–20. For SC1, the total electricity demand is expected to be about 59 GWh by 2020 and 100 GWh by 2030 (Fig. 8).

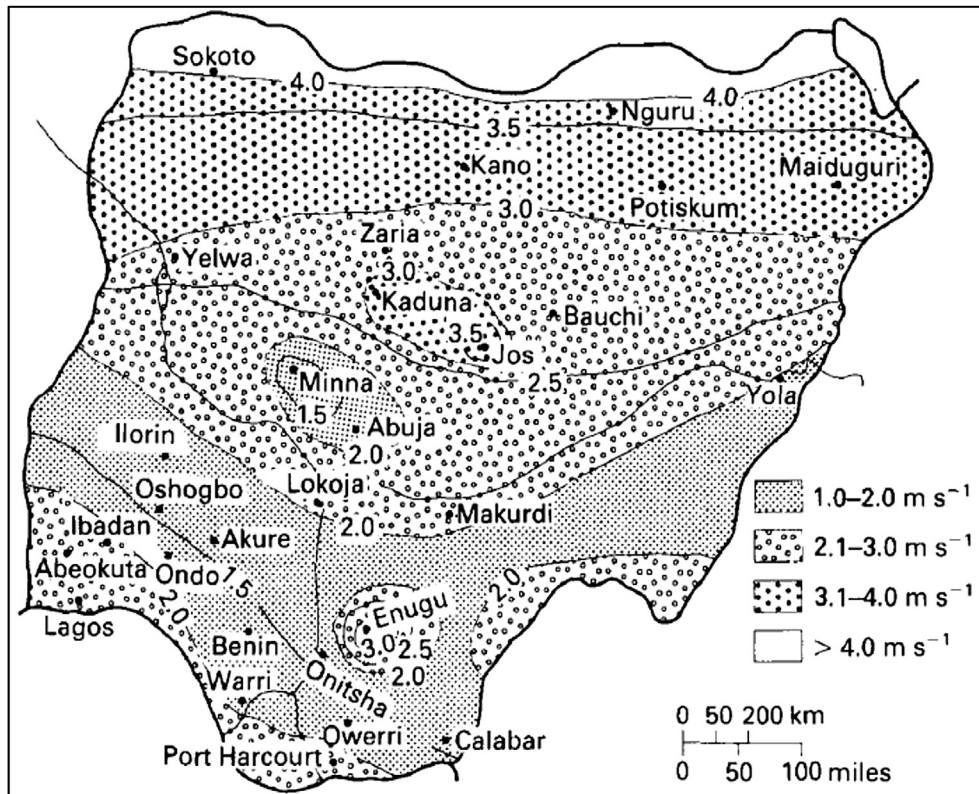


Fig. 7. Nigeria's annual wind speeds distribution (isovents at 10 m height) showing four different wind speed regimes [37].

In the base year of SC1, the total electricity generation output (Fig. 9) was 111.6 Million GWJ out of which 30% is from hydro fuel and the remainder is from the natural gas fuelled plants. This has improved in the reference scenario, where the total demand has increased dramatically considering a 6.8% growth rate in the peak load. By 2020, the total generation output increased to 247.7 Million GWh with coal, wind and solar fuels contributing 12.7% of the energy. With the introduction of 1 GW nuclear by 2022 and increasing the capacity of coal, solar and hydro and natural gas and nuclear, the generation output increased to 409.1 Million GWh in the end year.

The combined capacity (Fig. 10) has grown from 6.9 Thousand MW in 2010 to 32.6 Thousand MW in 2030. In the base year, only hydro and Gas accounted for the 6.9 Thousand MW of electricity by the end year of our simulation. Nuclear and the renewables (hydro, wind and solar) are expected to account for 42% of the total capacity by 2030.

The one hundred years GWP (global warming potential) for all GHGs (Fig. 11) due to emissions from the energy generation technologies (coal and natural gas) was calculated. In the base year of SC1 the total GWP was 7.9 million metric tons of CO₂ equivalent (MMTO-CO₂); due to the addition of the capacities of the technologies as well as increased in the demand for electricity the GWP for non-biogenic CO₂ has been increased from the base year to the end year. The GWP will triple (25.2 MMTO-CO₂) by 2030 with CO₂ contributing almost all the GWP.

For the SC2, the total electricity demand will be about 160 GWh by 2030 (Fig. 12), and going by the FGN's power sector reform plan, the country's generation capacity (Fig. 13) would grow from 6.9 to about 46 thousand MW by the end year. The generation output (Fig. 14) would grow six times the base year value natural gas and nuclear playing important role in the attainment of the electricity generation outputs. By 2030, the generation outputs of SC2 would

be 654.5 million GJ. The GWP of SC2 would doubled that of SC1 at its end year

By 2030, the GWP (of SC2, Fig. 15) would be 42.5 MMTO-CO₂, owing to the higher electricity demand in SC2 compared to the demand for SC1.

For the last scenario, SC3, the total electricity demand (Fig. 16) would be about 300 GWh and by 2030, the generation capacity (Fig. 17) is 86.7 thousand MW (16% – natural gas, 7% hydro, 40.5% –

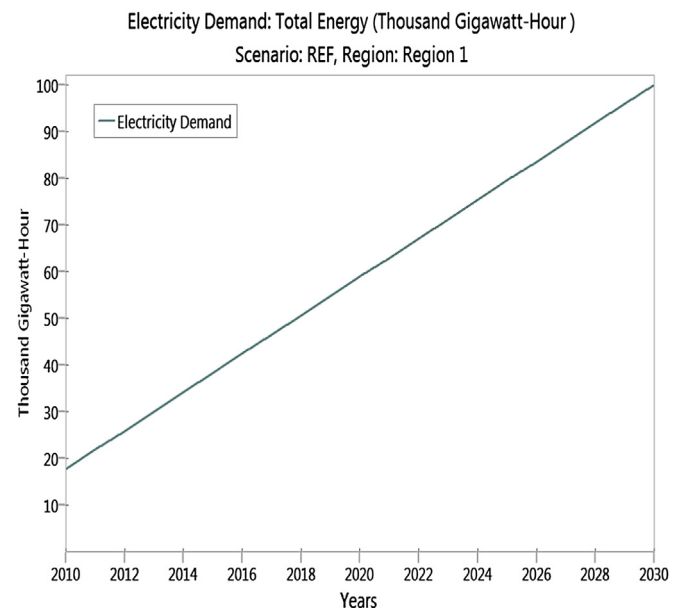


Fig. 8. The total electricity demand in Nigeria (SC1).

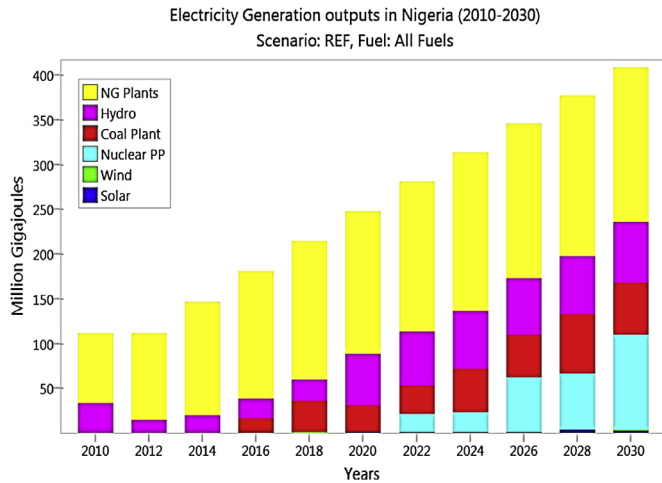


Fig. 9. Electricity generation outputs in Nigeria (2010–2030) contribution of fuels (SC1).

coal, 33% nuclear, 0.5% – wind and 3% – solar). By 2030, (Fig. 18) the generation outputs of SC3 would be 1.2 billion GJ and the corresponding GWP (Fig. 19) would be 80.2 MMTO-CO₂ this is trice and twice the end year values of SC1 and SC2, respectively.

6. A pioneer NPP (nuclear power plant) in Nigeria an insight of policy and safety issues

Nigeria plans to become a nuclear country by 2030 and has contacted the IAEA (International Atomic Energy Agency) for assistance in the deployment of its pioneer nuclear plant. According to the roadmap for the deployment of nuclear power plants for electricity generation in the Nigeria, the NAEC (Nigerian Atomic Energy Commission) plans to achieve at least 1000 MW in the next 10–12 years and the nuclear capacity is expected to grow up to 4000 MW by 2030 [32].

Since the Fukushima nuclear accident in Japan, nuclear energy has been under increased media scrutiny. Germany has implemented a radical transformation of the position of nuclear energy in the country, a position which is believed to have some political influence as the “post-Fukushima” German nuclear policy influences election results [42]. In the U.S, though there is no clear positions of the political parties on the expansion of U.S’s nuclear energy, a number of internal anti-nuclear activists like that of June

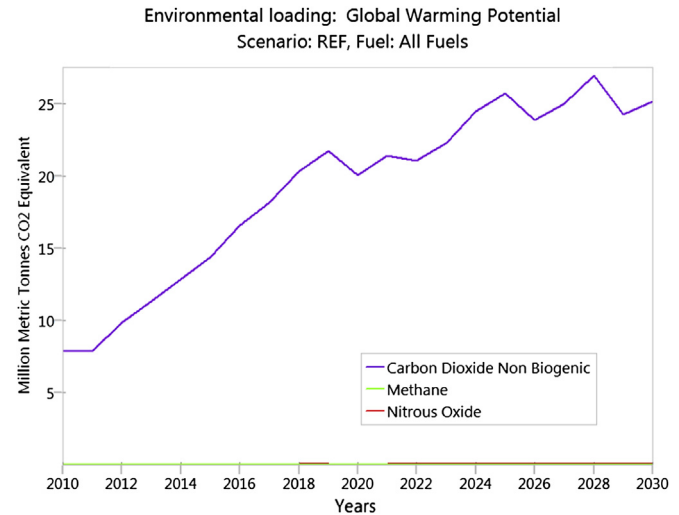


Fig. 11. The hundred years global warming potential based on the electricity generation in Nigeria for all GHGs (SC1).

2012 where three persons trespassed onto the Y-12 nuclear weapons plant in Oak Ridge, Tennessee to protest against the plant and a number of other anti-nuclear campaigns that have been reported in media.

Nuclear energy is currently the only energy technology with a secure base load electricity supply and no GHG emissions that has the potential to expand at a large-scale. A comparative study on the environmental impact and cost analysis of coal versus nuclear power in the U.S [14] showed that nuclear power industry has been over-regulated and the coal-burning power industry is under-regulated; and if the carbon-capture and sequestration tax is added, the levelized costs of coal and nuclear electricity production become comparable.

Contrary to the competitiveness of nuclear energy in U.S and European electricity markets, critic of the technology have highlighted some of the limitations of nuclear energy and argued that a more realistic analysis of the contribution of nuclear energy should mention that NPP can provide only base load power, while electric energy coming from hydropower and gas fired plants has a much higher value, as their production follows the demand variation for electricity [43]. Another issue that is considered in under scoring nuclear energy is the issue of nuclear waste generation, about which there has been ongoing debate among environmentalists on the geological disposal method of highly radioactive nuclear waste.

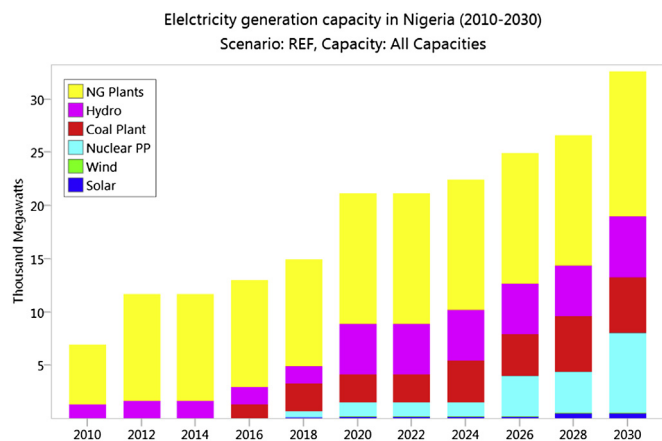


Fig. 10. The Total electricity generation capacity (2010–2030) of the different fuel in Nigeria (SC1).

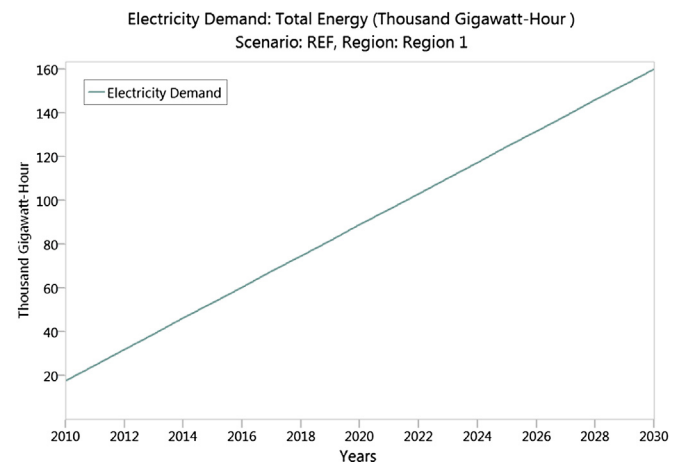


Fig. 12. The total electricity demand in Nigeria (SC2).

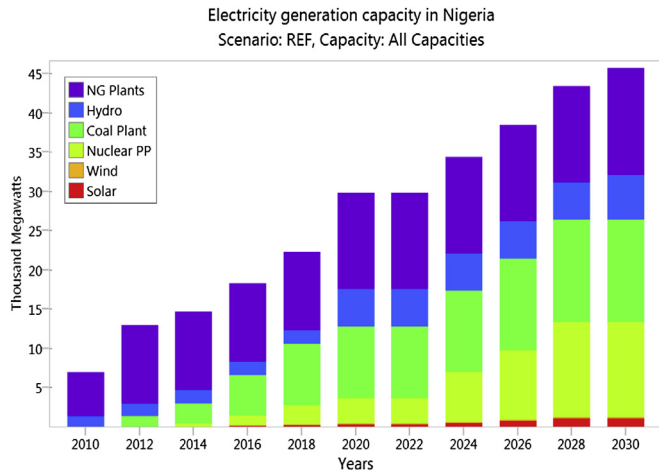


Fig. 13. The Total electricity generation capacity (2010–2030) of the different fuel in Nigeria (SC2).

The debate of nuclear energy replacing its fossil counterparts due the emission of GHGs which increases the thread of global warming has been on for a long period of time. Different countries have different political positions with regards to nuclear energy; for instance [42], mentioned that in the post-communist countries in Central Europe, there is no relevant political opposition to constructing NPPs whereas all political parties in Germany and Austria are strictly anti-nuclear.

On the other hand, the pro-nuclear have argued that it is competitive energy technology, e.g. in the US electricity market, and in recent years the performance of nuclear is similar to that of coal as depicted in Fig. 21.

Fig. 20 shows the U.S. electricity production cost in the period from 1995 to 2009, in 2009 cents per kWh. The production cost is defined as the sum of O&M costs and fuel costs. The percentage of fuel cost for nuclear power is only 28%, the cost of fuel for gas-powered plant is 89%, and for coal-burning plant 78% of overall production cost in 2009. The nuclear fuel cost consists of following components: the cost of conversion (4%), fabrication (8%), waste fund (15%), enrichment (31%), and uranium (42%).

Despite the problem connected to its cost and risk, nuclear energy is receiving increasing support from policy makers, who are

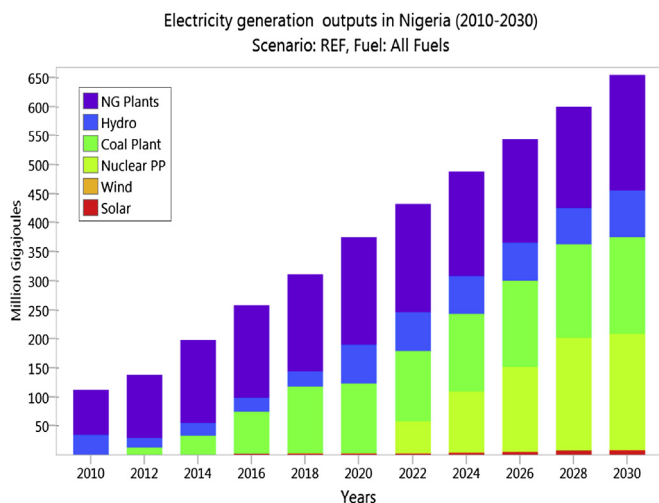


Fig. 14. Electricity generation outputs in Nigeria (2010–2030) contribution of fuels (SC2).

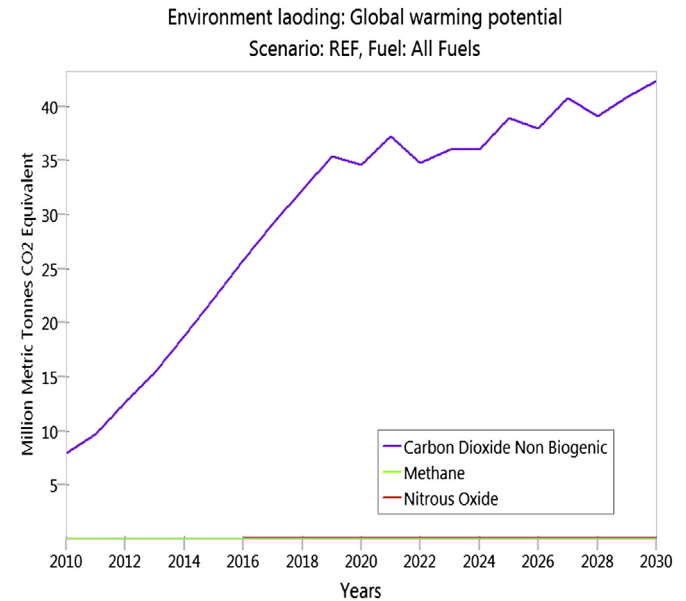


Fig. 15. The hundred years global warming potential based on the electricity generation in Nigeria for all GHGs (SC2).

willing to expand its use and some industry experts have called for the change in public perception of nuclear fuel [27,44]. To address the issue of CO₂ emission, policy makers around the world must realize that we will have a far better chance of reducing CO₂ emissions in time to curtail the incessant impact of global warming if we use both nuclear and renewable energy to replace fossil fuel energy [45].

In a recent study [32], we presented the state of nuclear power infrastructure of Nigeria. Describing how Nigeria is progressing towards nuclear power. The study concluded that the nuclear option has a significant potential to drive Nigeria transition to sustainable energy as long as environmental, social and economic issues are taken into consideration in its deployment.

In recent years, Nigeria is faced with number of insecurity issues ranging from the militancy activities in the coastal Delta to terrorism in some parts of the northern region. All efforts to address these security threats using military forces proved abortive, a situation that makes the government to reconsider its stands on not

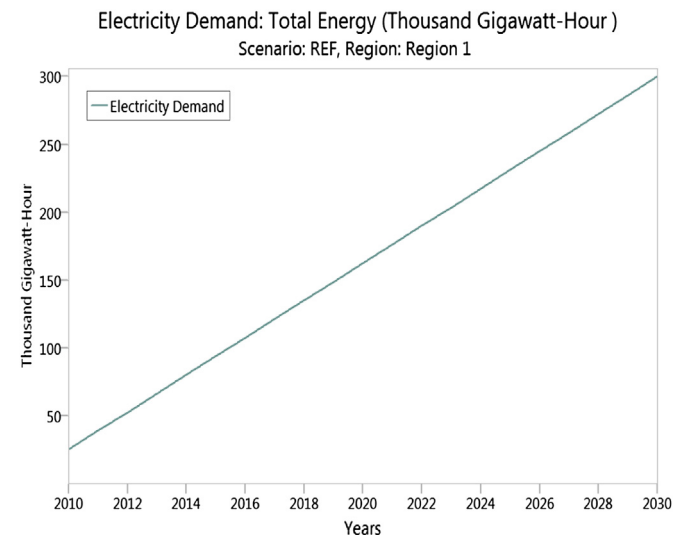


Fig. 16. The total electricity demand in Nigeria (SC3).

Electricity generation capacity in Nigeria (2010–2030)
Scenario: Reference

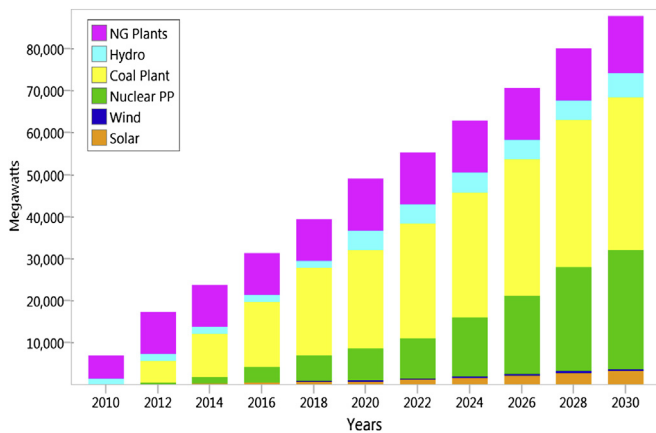


Fig. 17. The total electricity generation capacity (2010–2030) of the different fuel in Nigeria (SC3).

dialoguing with armed groups and begin to consider a round table dialogue with these groups. These groups may disrupt the development of NPP program by holding workers to ransom as it is witnessed in the oil producing region. Such domestic activists could result in construction delays, cost overrun stories as well as increasing the cost of building the NPP [46]. Another sensitive issue the government must take very seriously is the possibilities of a domestic terrorist attack on nuclear facilities and the possibility of diversion of nuclear materials by connected terrorist network for weapon making. It is important for all plans for new nuclear program to recognize the increased need for the protection of nuclear facilities against external activities such as terrorist attack and the possible diversion of radioactive materials for the proliferation of nuclear weapons.

Despite some progress have being achieved in some areas of Nigeria nuclear program, including the ratification of international treaties, development of regulatory agencies and signing of bilateral technical cooperation agreements, great challenges remain: a low-grade grid, underdeveloped electricity market, lack of skilled manpower, widespread corruption and a dubious history of success in large, government-managed projects, render the proposed NAEC timeline for Nigeria's pioneer NPP and expansion plans almost unrealistic [47].

Energy Generation outputs in Nigeria (2010–2030)
Scenario: REF, Fuel: All Fuels

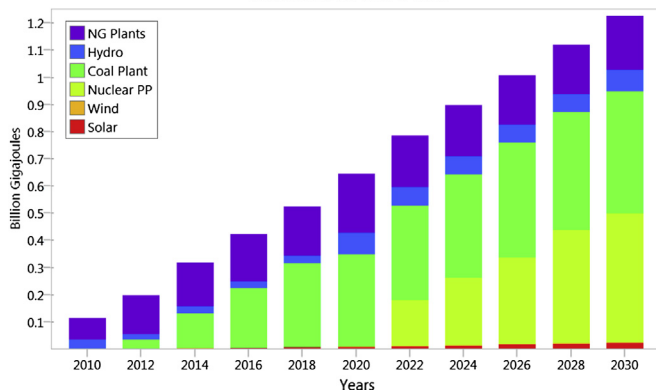


Fig. 18. Electricity generation outputs in Nigeria (2010–2030) contribution of fuels (SC3).

Environmental Loading : Global warming potential due to electricity generation in Nigeria
Scenario: Reference

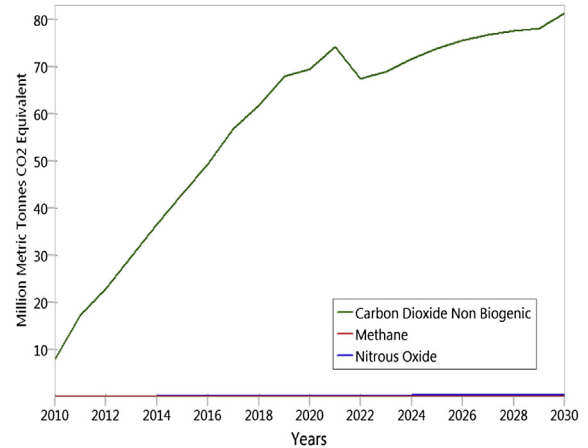


Fig. 19. The hundred years global warming potential based on the electricity generation in Nigeria for all GHGs (SC3).

6.1. Atmospheric dispersion modelling at Nigeria's first NPP

The atmosphere is an important pathway for the transportation and deposition of radioactive release from NPP thereby to man and the environment. The health consequences of any nuclear accident depends on wind speed, wind direction, rainout and atmospheric stability, which are vital meteorological parameters considered in the siting of nuclear power installations [48]. The regulatory standard set by the IAEA is to calculate the downwind ground deposition and the air concentration of the released effluent using atmospheric dispersion models [49].

Computer models are now an important part of the environmental health and safety assessment. The study and improvement of techniques in atmospheric dispersion modelling of radioactive effluent in risk assessment and emergency response date back to half a century ago [50,51] and one of the most conventional techniques of atmospheric dispersion modelling of radioactive effluent is the GPM (Gaussian Plume Model), which depicts the approximate distribution of near ground concentration over time. But for the purpose of emergency response and risk assessment some advanced models (such as Langragian puff and particle models) are currently favoured.

Modelling of nuclear emergency may not be well satisfied by the existing models in the case of actual emergencies as the current models are not well designed for complex terrain and urban environments (Yao, 2011). However, the model of the US EPA (Environmental Protection Agency); AERMOD-model has been used for accurate dispersion calculation of radioactive fallouts by Ref. [52];

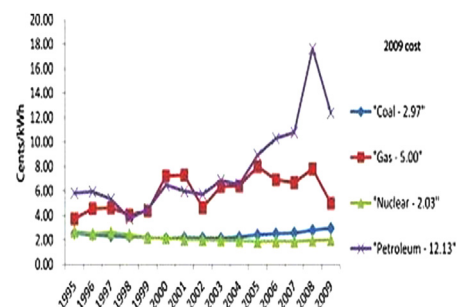


Fig. 20. Cost of electricity generation in the US in constant 2009 cent/kwh [14].

and has been considered by experts in NPP safety as a candidate for offsite doses calculations [53].

6.2. Methodology for atmospheric releases impact assessment

The AERMOD modelling system consists of two pre-processors and the dispersion model. The AERMIC meteorological pre-processor (AERMET) provides AERMOD with the meteorological information it needs to characterize the PBL (planetary boundary layer) and the terrain pre-processor (AERMAP) which characterizes the terrain, and generates receptor grids for the dispersion model (AERMOD). All theories and calculations used in the generation of the AERMOD source code are available in literature [54–57].

Fig. 21 shows the data flow and processing information in AERMOD showing the two pre-processors (AERMET and AERMAP) with their functionalities.

The radiation dose was calculated analytically from the average air concentration and ground contamination data obtained from the AERMOD run.

The proposed site for the NPP is located at Itu Akwa Ibom state Nigeria (5.20N and 7.960E). For the purpose of this study, the coordinate was changed into UTM coordinate system using the WGS 84 map datum. The area is located in the UTM zone 32. A Pseudo nuclear power plant with similar engineering and technical specifications was adopted in the simulations.

Since the site is located in rural area and the data quality for proximate observation stations are of the order of 10%, the surface and upper air meteorological data needed by AERMOD were generated using MM5 mesoscale modeling system provided by the PSU/NCAR (Pennsylvania State University/National Centre for Atmospheric Research) <http://www.mmm.ucar.edu/mm5/>. The MM5 generated data were processed using the AERMET code in order to prepare the hourly meteorological files required by AERMOD. The terrain information needed by AERMAP was provided in the form of 7.5 min 100 × 100 km DEM (digital elevation model) data. The DEM data was processed using the AERMAP pre-processor. The EPZ (emergency planning zone) of the proposed facility is within a 16 km radius, with the facility at the center (Fig. 22).

Fig. 23 presents the direction, from which the wind blew from in the study area for the period of 1st January to 31st December, 2011. Greater percentage of the wind blew from the S and the SSW directions with the speed ranging from 3 to 6 m/s (Fig. 23).

The precipitation rate plays important role in the removal of pollutant from the air (wet deposition), Fig. 24 is a time series plot for every hour (in 2011) showing the variation of the temperature and precipitation rate. The maximum temperatures were in April and May, the temperature was between 290 and 300 K. The highest precipitation rate of about 20 mm/h was witnessed in the mid of

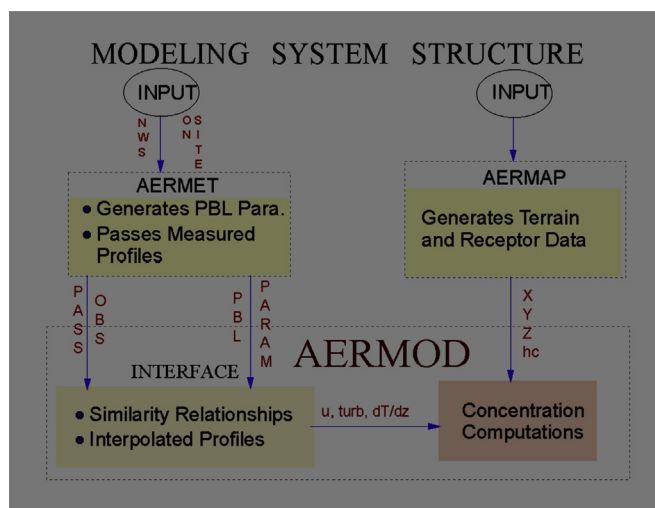


Fig. 21. Data flow in AERMOD (US EPA, 2004) [58].

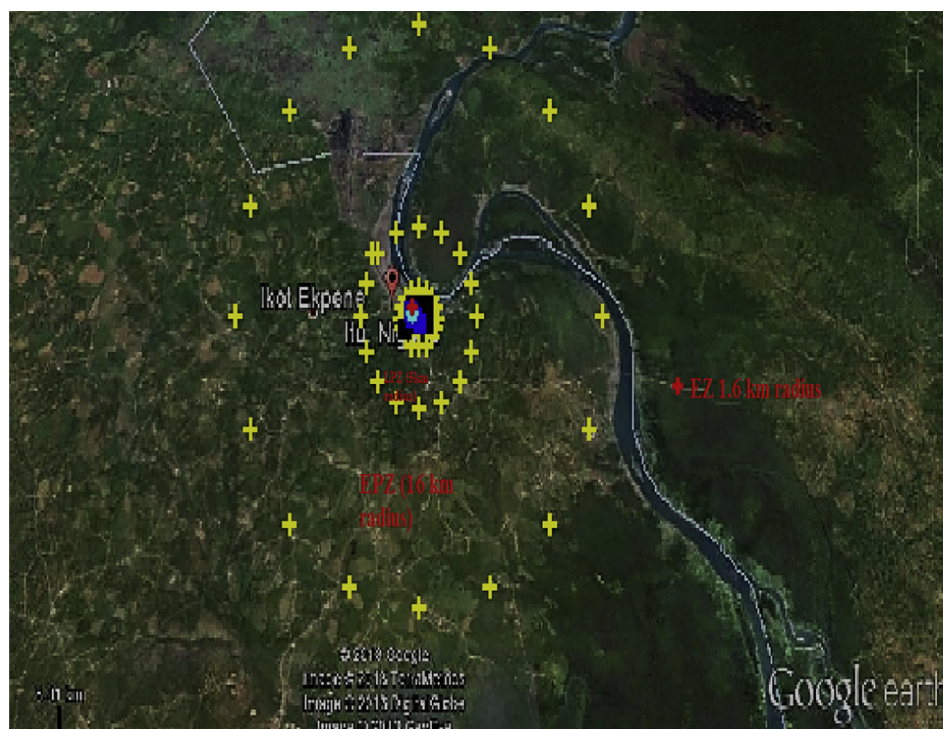


Fig. 22. The modelling (Emergency Planning zone) domain with a pseudo NPP.

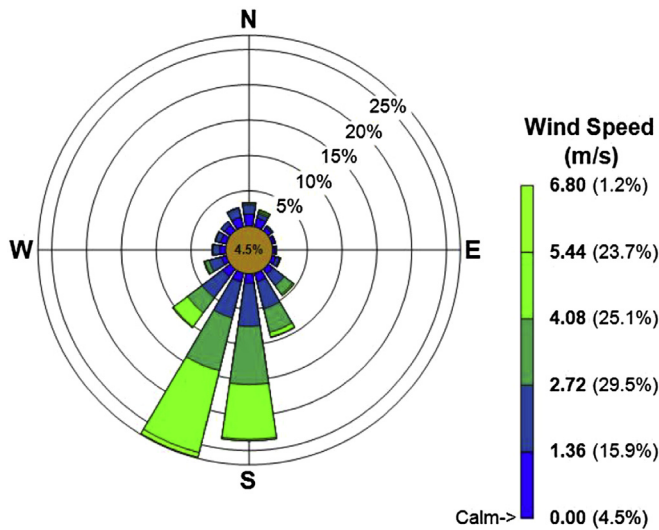


Fig. 23. Windrose plot of the NPP site.

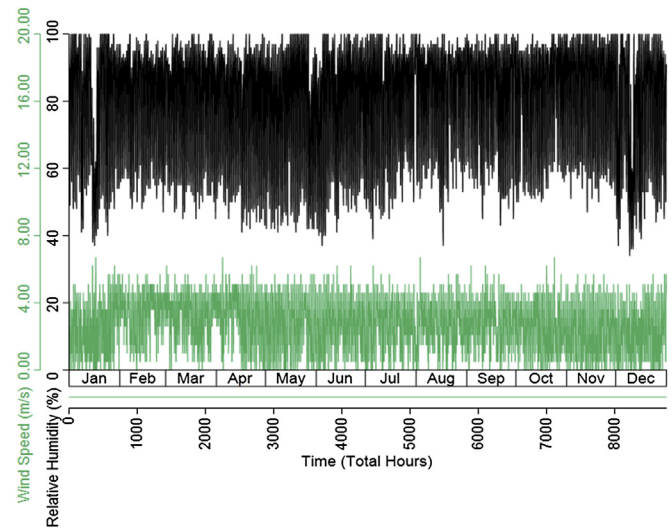


Fig. 25. Wind speed (m/s) and relative humidity (%).

September, 2011, whereas the months of January and December were relatively dried.

The highest wind speeds were experienced in mid-January, April and October and December endings. As for the relative humidity, it varies from about 40% to 60% with peaks in January, June, August and December. This information is depicted in Fig. 25.

We assumed that all emissions were from the stacks of the NPP and the stack parameters used in the simulation with AERMOD are presented in Table 8.

6.3. Normal operations releases

Small amount of ^{14}C (carbon-14) are generated during the routine operations of NPPs, this is due to neutron capture in LWR (Light Water Reactor). The isotope is released in two forms; oxides i.e. $^{14}\text{CO}_2$ and in reduced form; $^{14}\text{CH}_4$ which is subsequently absorbed by plants and transferred to animals. Man ingests ^{14}C via fixation of $^{14}\text{CO}_2$ by green plants, which subsequently are

consumed directly and indirectly via consumption of milk and meat from animals [59].

In the dispersion calculations, ^{14}C was assumed to be released continuously during the year in the form of $^{14}\text{CO}_2$ and in the dose calculations of the deposited particles, the resuspensions of the respirable particles was considered, since radioactive materials that are deposited on to the ground are resuspended either by wind or by disturbance [60]. The adopted normalized release rate of ^{14}C in the simulation is $6 \times 10^3 \text{ Bq s}^{-1}$ per GW (e) and the annual air concentration and deposition of the radionuclide were calculated. The contour plot (Fig. 26) of the ground level concentration of ^{14}C shows that the intensity is higher within the NPP facility and the contour extends to the facility's fence (almost 1 km away).

The maximum (first high) GLC for ^{14}C is $0.01 \times 10^{-3} \text{ kBq/m}^3$ (micro Becquerel per cubic meter) and the minimum 28.0864 $\mu\text{Bq/m}^3$. [Eq.(1)] was used to calculate the radiation dose as follows, assuming a 2000 work-hour per annum (for a plant staff) and the $\text{CF}_{\text{immersion}}$ for ^{14}C is 8.06×10^{-10} .

$$E_{\text{immersion}} = C \times \text{CF}_{\text{immersion}} \times t \quad (1)$$

where

$E_{\text{immersion}}$ = Effective dose from external exposure due to immersion in contaminated air [mSv].

C = Average concentration of radionuclide in air [kBq/m^3].

$\text{CF}_{\text{immersion}}$ = Conversion factor for radionuclide [$(\text{mSv/h})/(\text{kBq/m}^3)$].

t = Exposure duration [h].

Then the annual radiation dose for the plant worker is $0.16 \times 10^{-10} \text{ mSv}$ (milli Sievert) which is quite small. This value

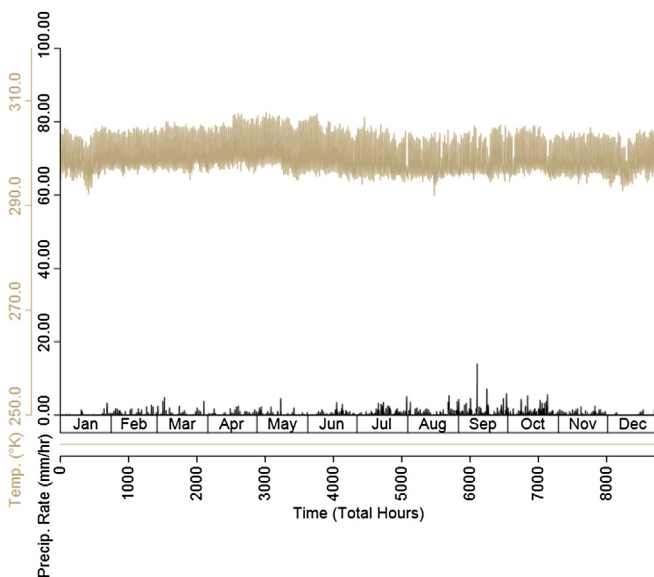


Fig. 24. Time series plot of the precipitation rate (mm/hr) and temperature (°C).

Table 8

Stack information.

Release velocity	15 ms^{-1}
Release rate	Varies for each radionuclide ^a
Stack height	50 m
Stack temperature	300 K
Stack diameter	4 m

^a Depend on the pollutant (radionuclide).

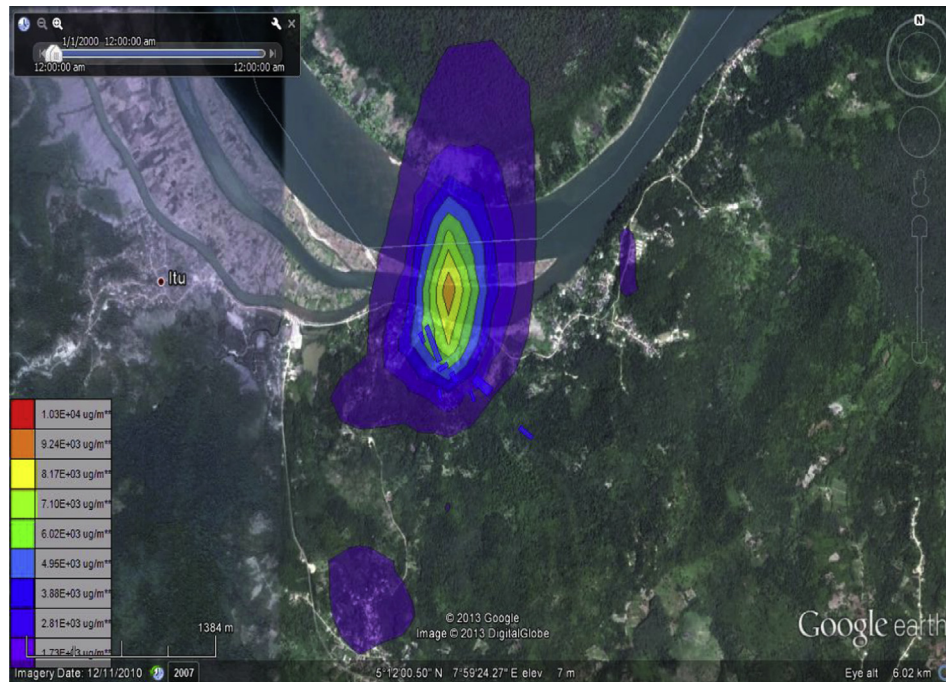


Fig. 26. Contour plot of the GLC (ground level concentration) of ^{14}C .

should not be confused with the total radiation dose of the worker as this accounts for only one radionuclide out of the many others which are released from NPPs during routine operations. It is important to mention here that the routine operation releases of NPPs are highly regulated and to investigate the risk of NPP's siting on the environment one must consider a hypothetical accident situation where the release rates are of 3 order of magnitude than the routine operations and volatile radionuclide with higher cancer risk (like Caesium-137 and iodine-131) are involved. This type of dispersion calculations are useful in planning for emergency situations involving NPPs.

7. Conclusion

This paper presents an overview of the electricity crisis in Nigeria, the policy issues and environmental ramifications of the power sector reform act as well as dispersion modeling of emissions from Nigeria's (yet to be constructed) pioneer NPP. The over-dependence of the Nigerian energy sector on petroleum has slowed down the development of alternative fuels. In order to achieve the Vision 20:2020, efforts must be made toward achieving a diversified energy supply mix, which will ensure greater energy security for Nigeria. However, in its desperate attempt to address the energy poverty, the Government may consider solely the further development of conventional electricity technologies (like coal, oil and gas) that are readily available in Nigeria with little or no concern on the environmental impact of these technologies. The GWP of Nigeria's power sector expansion plan has been estimated considering three different scenarios which consider the growth rate of the peak load using the LEAP model. The GWP is only for the electricity generation sector as it does not account for emissions from the industrial, residential and oil and gas sectors. Although the funds required for the power sector expansion are enormous, so is its environmental ramifications, as it involves destruction of ecosystem, deforestation as well as a long term emissions of CO_2 and other toxic substance into the biosphere.

Future Task

The future task will involve the use of state of the art modelling tools for detailed environmental impact assessment (ERICA tool) and human health impact assessment of the NPP using global scale dispersion model (like HYSPLIT 4) and GENII codes. It is hoped that the results of this task will help policy makers in decision making with regards to the NPP's construction and operation.

Acknowledgement

Our sincere appreciation goes to the Malaysian Ministry of Higher education and the Universiti Teknologi Malaysia for providing a research grant (QJ130000.2526.03H67) from which part of this work is supported.

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